

DepthVista_Helix

DepthVista Console Application User Manual



e-con Systems®
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e-con Systems

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Introduction to DepthVista

- DepthVista_Helix is a 3D camera based on Smart Indirect Time of Flight (iTOF) technology, from e-con Systems, a company with over two decades of experience in designing, developing, and manufacturing OEM cameras.
- DepthVista uses a 1/3.2" AF0130 CMOS sensor from Onsemi, designed for precise and high-resolution 3D depth measurement, making it ideal for real-time 3D applications.

Prerequisites

The prerequisites are as follows:

- DepthVista console application source code
- CMake (version 3.5 and above)
- DepthVistaSDK
- OpenCV library (version 4.5.4 and above)

Description

- DepthVista is a 3D depth camera based on Onsemi's Smart iTOF sensor, featuring **on-camera depth processing** for accurate 3D measurements without the need for external processing.
- The **1.2 MP** CMOS Smart iToF Sensor, with Advanced 3.5 mm Pixel Stacked BSI Technology offers **Superior Low-Light** and **Ambient-Light Performance**.
- It can provide high-resolution 1.2MP depth output for a range of 0.2m to 2m. With its **Dual Laser operation**, it also delivers **high-accuracy** VGA depth output for a range of **0.5 m to 6 m**.

Note: The effective depth range may be reduced under outdoor operating conditions due to ambient sunlight interference. In extreme sunlight environments, the measurable depth range can decrease to as low as 1m in HD mode and 3m in VGA mode.

- Prioritizing safety and reliability, it features Laser **Eye Safety Monitoring**.
- Designed to withstand tough environments, the GMSL camera comes with an IP67 - rated enclosure, making it durable and versatile.
- It comes in USB and GMSL variants, with GMSL variant supporting NVIDIA Jetson AGX Orin NX/Nano development kits, enabling easy evaluation and integration into applications.

Installing DepthVistaSDK in Linux

This section describes the installation of DepthVistaSDK which is essential for building DepthVista Application.

The steps to install the DepthVistaSDK are as follows:

1. Run the following command to extract the **package** file.

```
Unzip -X <packageName.zip>
```

<Extracted Directory>\Linux\Bin\Ubuntu22.04\ARM\SDK will have an install.sh file.

Note:

For x64 platform the install.sh file will be present in in <Extracted Directory>\linux\Bin\Ubuntu22.04\x64\SDK\DepthVistaSDKInstaller

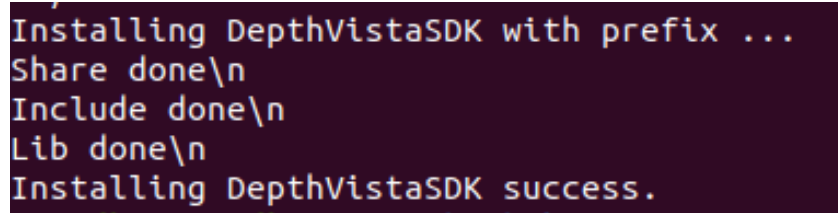
2. Open the folder containing install.sh in terminal.
3. Run the following command to give executable permission for install.sh file.

```
chmod +x install.sh
```

4. Run the following command to install the **DepthVistaSDK**.

```
sudo ./install.sh
```

Once installation is successful, **Installation DepthVistaSDK success** message appears.



```
Installing DepthVistaSDK with prefix ...  
Share done\  
Include done\  
Lib done\  
Installing DepthVistaSDK success.
```

Installing DepthVista Application

This section describes how to install the DepthVista Console application.

Step 1-Welcome Screen

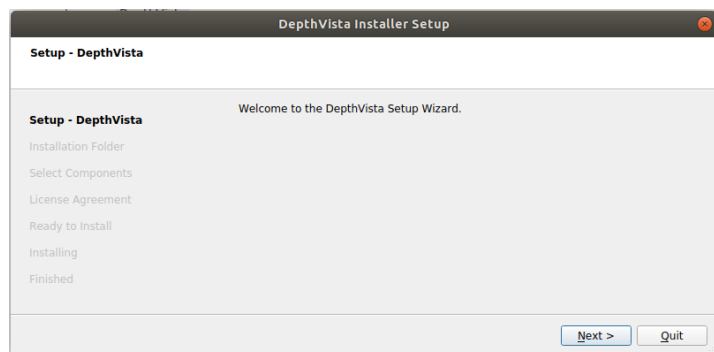
As a first step, extract the given package using the following command.

```
unzip -X <packageName.zip>
```

<ExtractedPath>/DepthVista_Application_Package_Ver_x.x.x.x/Linux/Bin/Ubuntu22.04/ARM will have **DepthVistaViewer_ARM_x.x.x.x_xxxxxxxx.run** file. Run the **DepthVistaViewer_ARM_x.x.x.x_xxxxxxxx.run** file.

Note: For x64 platform

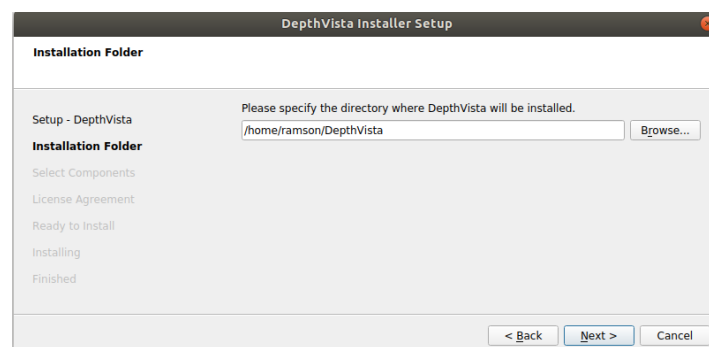
When you run the **DepthVistaViewer_ARM_x.x.x.x_xxxxxxxx.run** file, the Welcome screen appears as shown below.



Click the **Next** button.

Step 2-Installation Path Selection

On clicking the **Next** button, you can view the screen like the installation path selection window as shown below.



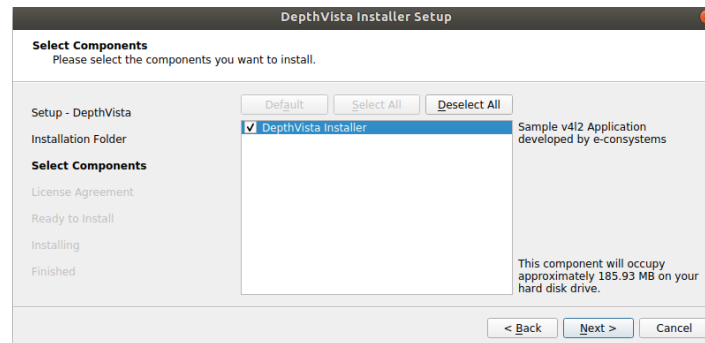
You can either use the default installation path **home/user/DepthVista** or change to another installation path by clicking the **Browse** button.

Click the **Next** button.

Step 3-Component Selection

On clicking the **Next** button, the components selection window will appear.

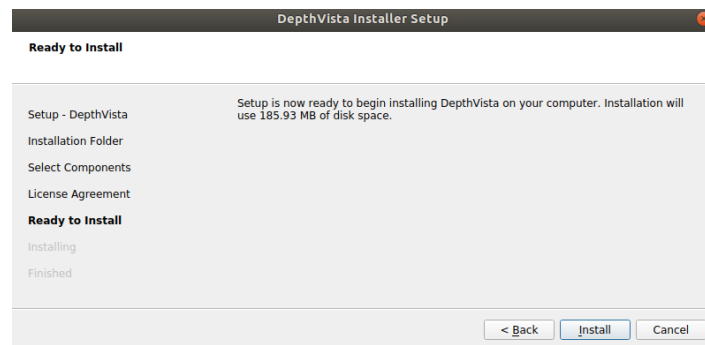
Make sure to select the **DepthVista Installer** as shown below.



After selection, click the **Next** button.

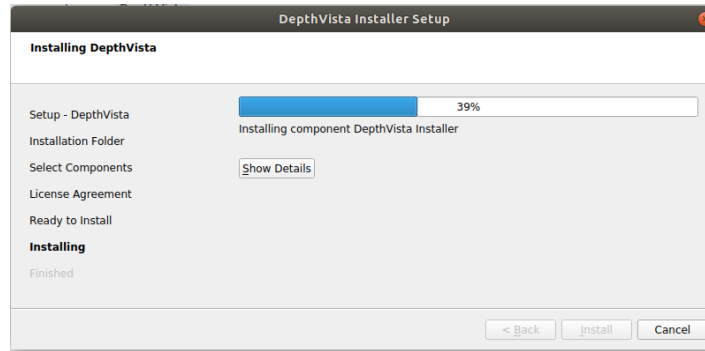
Step 4 - Installation Process

On clicking the **Next** button, the ready to install window appears as shown below.



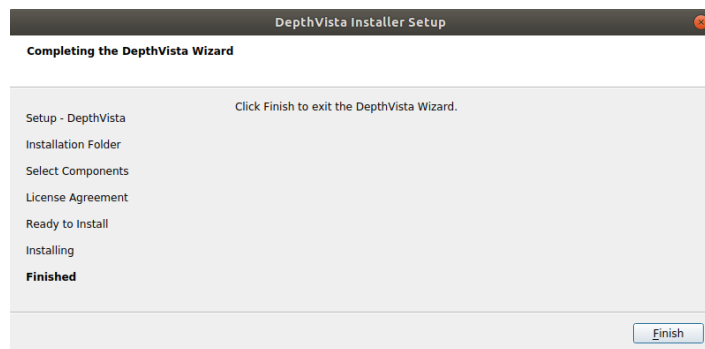
Click the **Install** button to install the DepthVista application.

Now, all the required files are extracted and copied to specific location. The progress of installation appears as shown.



Step 5 - Installation Complete

After the DepthVista installation, you can view the DepthVista setup completion window as shown below.



Click the **Finish** button to exit the setup wizard after the completion of installation.

Using DepthVista Console Application

This section describes how to use the DepthVista Console application.

Launching DepthVista Console Application in Ubuntu

The steps to launch DepthVista Console Application in Ubuntu are as follows:

1. Run the following command from the directory where DepthVista Application is installed to run the application.

```
sudo ./DepthVistaConsoleApp
```

Device Selection

After starting the application user can see the list of the available devices.

```
e-con's Sample Application for DepthVista
Demonstrates the working of e-con's DepthVistaSDK

DepthVista SDK-Version = 1.0.14

Number of Camera Devices Connected to the Port : 1
Camera Devices Connected to the PC Port :

0 - Exit
1 - See3CAM_TOF_CU13 (110CBF61 \\?\USB#VID_2560&PID_C11C&MI_00#6&110CBF61&0&0000#{65E8773D-8F56-11D0-A3B9-00A0C9223196}\GLOBAL)

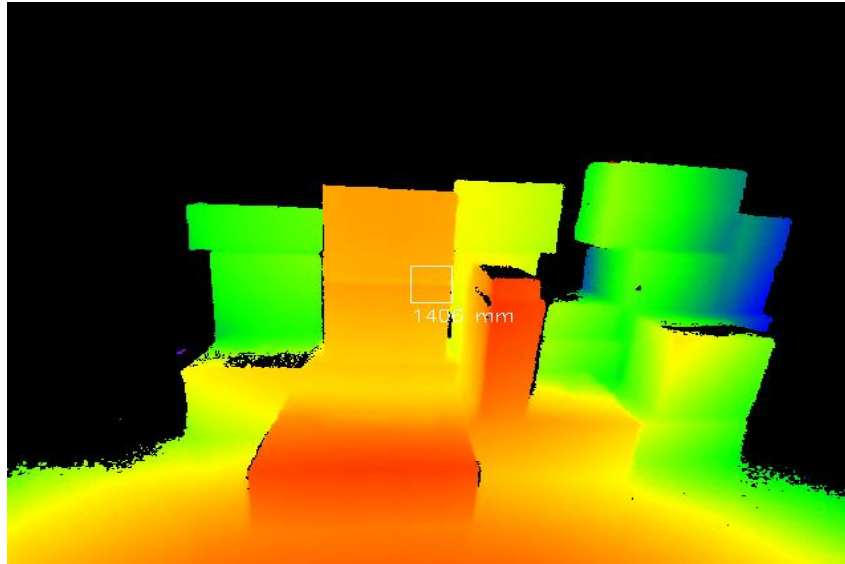
Pick a Camera Device to Explore :
```

After selecting the device to explore stream will start from the selected device, and the camera properties will display

The IR Preview will be shown as below:



Depth Color Map preview will shown as below:



Camera Properties

Camera properties that can be explored will be listed as shown below.

```

0    - Exit
1    - Back
2    - Streaming Mode
3    - Capture Frames
4    - Depth Denoise
5    - Confidence Threshold
6    - Integration Time
7    - TOF IR Gain
8    - Flying Pixel Filter
9    - Get Temperature data
10   - Post Processing
11   - Confidence Mode
12   - AVG Depth
13   - Unique ID
14   - Read Firmware Version
15   - Serial Number
16   - Laser Safety
17   - FPS Control

Pick a Relevant Choice of Camera Properties :
```

Streaming Mode

After selecting the **Streaming Mode** option, all the streaming modes supported by the device will be listed as shown below.

```
Total Number of Streaming Modes Supported by the Camera:      2
    0 - Exit
    1 - Back
    2 - Main Menu
    3 - VGA_LongRange_6m
    4 - 1.2MP_ShortRange_2m

Pick a Relevant Streaming Mode:
```

After selecting the preferred **Streaming Mode**, user can view the preview based on the selected streaming mode.

Here, selecting the VGA_LongRange_6m data mode will display the Depth Colormap and IR frame in 640x480 Resolution. Whereas selecting the 1.2MP_ShortRange_2m will display the Depth Colormap and IR frames in 1280x960 Resolution.

Note:

Streaming Limitations

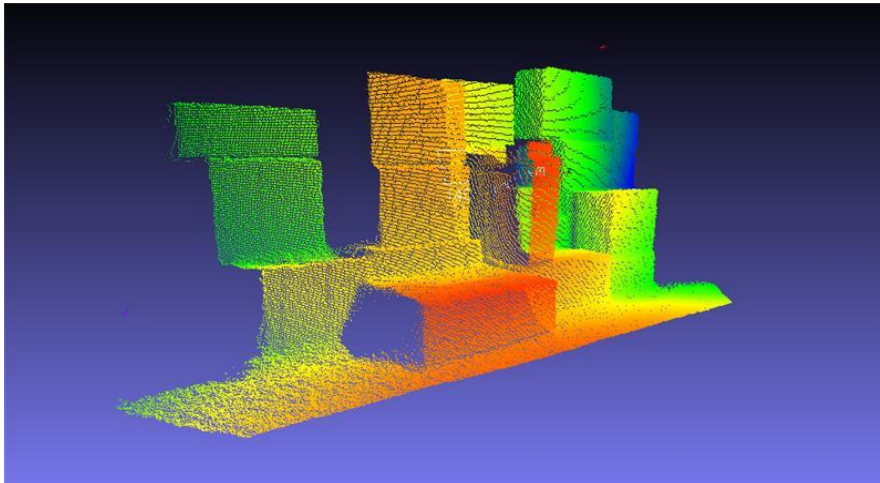
- **When streaming in VGA mode, the console app can reliably deliver frames at the expected 30 or 50 FPS.**
- **In HD mode (1.2MP resolution), frame drops may occur due to the higher data throughput required. The extent of this drop depends on the host system's configuration and processing capability.**
- **The number of devices simultaneously streaming in HD mode further impacts performance, contributing to additional FPS reduction.**

Capture Frames

After selecting **Capture Frames** images will be saved based on the streaming mode and the location of the images will be shown as follows.

```
DepthColorMap frame is successfully saved as DepthVistaCam0_Depth_2026_5_20_15_49_5.bmp
Raw Depth frame is successfully saved as DepthVistaCam0_raw_2026_5_20_15_49_5.raw
confidence frame is successfully saved as DepthVistaCam0_confidence_2026_5_20_15_49_5.raw
3D PLY is saved as DepthVistaCam0_PLY_2026_5_20_15_49_5.ply
IR frame is successfully saved as DepthVistaCam0_IR_2026_5_20_15_49_5.png
Raw IR frame is successfully saved as DepthVistaCam0_IR_raw_2026_5_20_15_49_5.raw
```

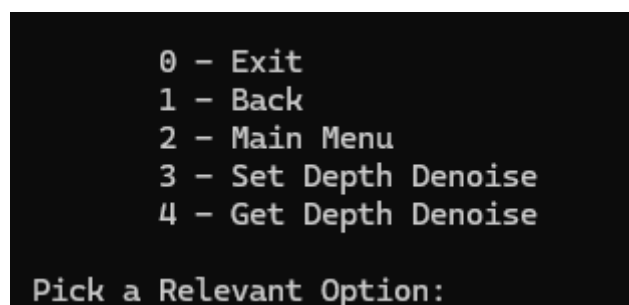
The PLY file will be displayed as below:



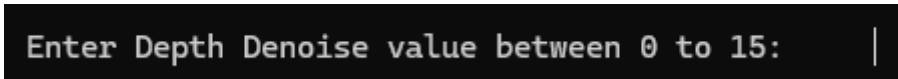
Depth DeNoise

- The Depth Denoise filter performs bilateral denoising by analysing the spatial neighbourhood of each pixel.
- It averages all valid and similar neighbouring pixels, selected based on the given denoise threshold, using spatial weights to filter outliers and to preserve edges

After selecting the **Depth Noise** option user can see the options as shown below.



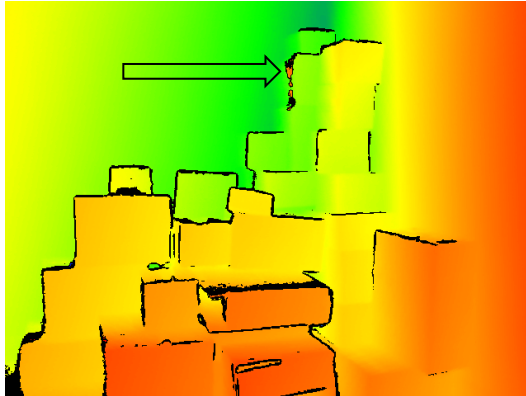
1. **Set Depth Noise** user should enter the value between 0 to 15.



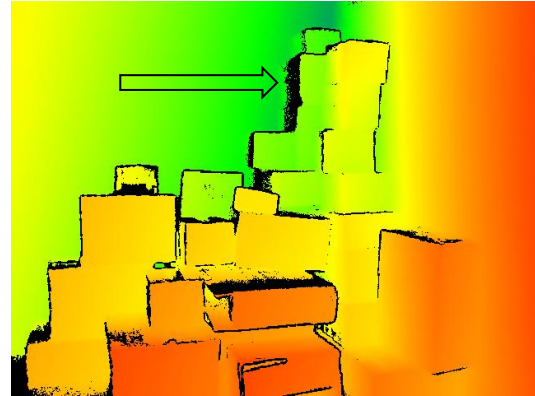
```

Enter Depth Denoise value between 0 to 15: |
  
```

Example : Setting the **Depth Noise** parameter:



Depth Denoise Value : 7



Depth Denoise Value : 15

2. **Get Depth Noise** used to get the current depth noise value.

```
Pick a Relevant Option:      4
Depth Denoise value of the camera 0 : 12
```

Confidence Threshold

- Confidence data provides an estimate of the precision of the depth pixels.
- A higher confidence value means the depth measurement is more likely to be accurate, while a lower value indicates uncertainty or noise.
- Using confidence threshold, less reliable depth pixels which are below a particular confidence value can be filtered out.

After selecting the **Set Confidence Threshold** user can see the options as shown below.

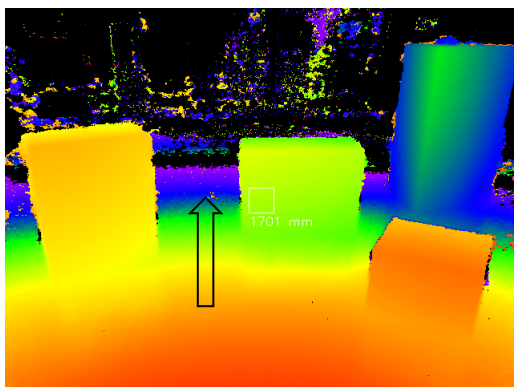
```
0 - Exit
1 - Back
2 - Main Menu
3 - Set Confidence Threshold
4 - Get Confidence Threshold

Pick a Relevant Option:
```

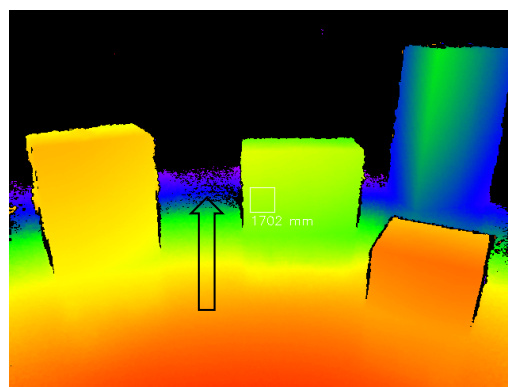
1. **Set Confidence Threshold** user should enter the value between 0 to 500.

Enter Confidence Threshold value between 0 to 500:

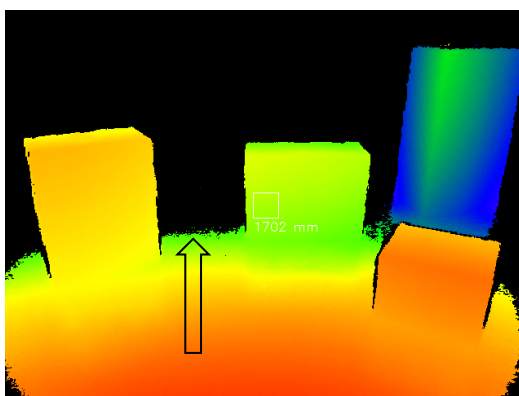
Example : Setting the **Confidence Threshold** parameter:



Confidence Threshold value : 0



Confidence Threshold value : 30



Confidence Threshold value : 50

2. **Get Confidence Threshold** used to get the current confidence threshold value.

Pick a Relevant Option: 4
Confidence Threshold value of the camera 0 : 180

Integration Time

- The integration time is the sensor's exposure period per phase, expressed in microseconds.

- Higher integration time improves the signal-to-noise ratio (SNR), enabling farther operating ranges and reduced depth variance, at the cost of increased power consumption.
- Lower integration times are typically sufficient for shorter range modes. Increasing exposure time also helps in obtaining more accurate depth for black or less reflective objects.

After selecting the **Set Integration Time** user can see the values as shown below.

```

0 - Exit
1 - Back
2 - Main Menu
3 - Set Intgeration Time
4 - Get Intgeration Time

Pick a Relevant Option:

```

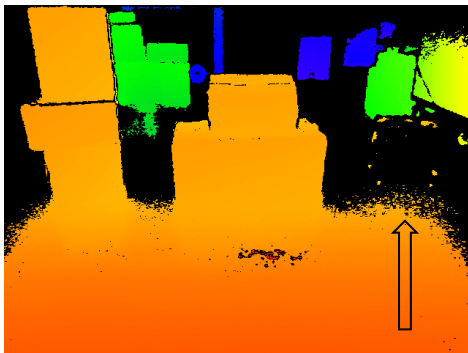
1. **Set Integration Time** user should enter the value between 50 to 800.

```

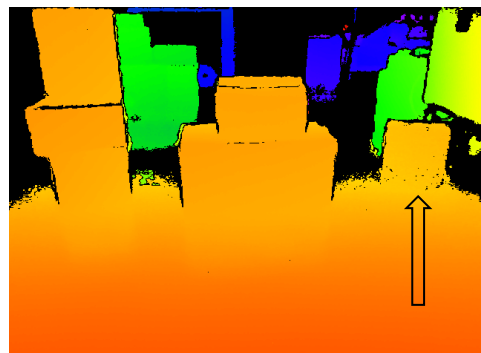
Enter Integration time value between 50 to 800:

```

Example : Setting the **Integration time** parameter:



Integration Time : 150



Integration Time : 250

In VGA 50FPS mode, the integration time is limited to 270us, thereby the effective range may be reduced.

```

Enter Integration time value between 50 to 270:

```


2. **Get Integration Time** used to get the current integration time.

```
Pick a Relevant Option:      4
Integration time value of the camera 0 : 183
```

TOF IR Gain

The IR gain can be increased or decreased to control the brightness of the IR image. It is a software gain and does not have any effects on the IR value of the RAW frame from the sensor.

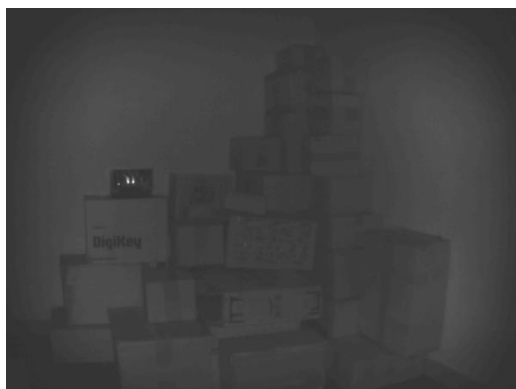
```
0 - Exit
1 - Back
2 - Main Menu
3 - Set IR Gain
4 - Get IR Gain

Pick a Relevant Option:
```

1. **Set IR Gain** used to set IR Gain value. User should enter the value between 1 to 100.

```
Pick a Relevant IR Gain between 1 to 100:
```

Example : Setting the **IR Gain** parameter:



IR Gain value : 16



IR Gain value : 50

2. **Get IR Gain** used to get the current IR Gain value

```
Pick a Relevant Option:          4
Current IR Gain value is : 13
```

Flying Pixel Filter

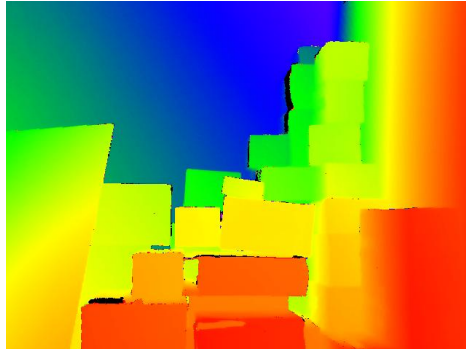
- Flying pixels are false pixels which are non-existent but exist in 3D point cloud. The lateral resolution of TOF sensors is generally less.
- Hence one pixel covers a large area in 3D space. If this area has objects at different depths, they combine into a single pixel causing flying pixels.
- This effect is more prominent at the boundaries of the objects. This is because at the edges, light from foreground and background may combine to produce intermediate erroneous depth values, which can be seen as a wave of pixels known as flying pixels. If a pixel has large depth discontinuity from its neighboring pixels, it is considered as flying pixel. This pixel will be termed as invalid and ignored in 3D point cloud.

After selecting the **Flying Pixel Filter** user can see the values as shown below.

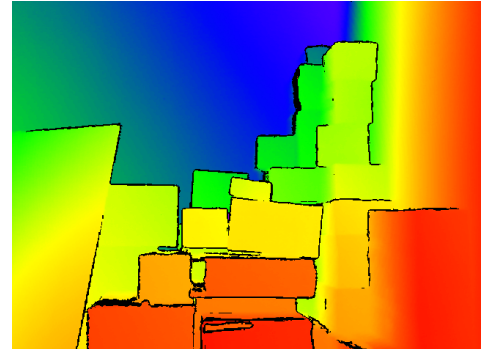
```
Pick a Relevant Choice of Camera Properties : 8

0 - Exit
1 - Back
2 - Main Menu
3 - Flying Pixel filter OFF
4 - Flying Pixel filter ON
```

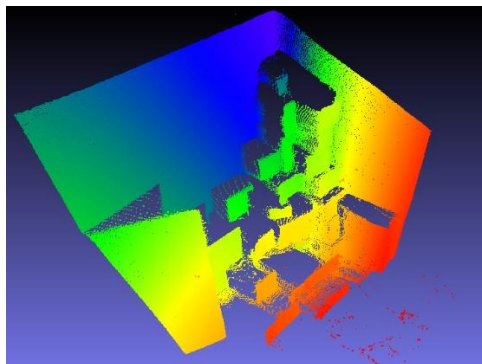
- **Flying Pixel filter OFF** – used to disable the Flying pixel filter.
- **Flying Pixel filter ON** – used to Enable the Flying pixel filter.



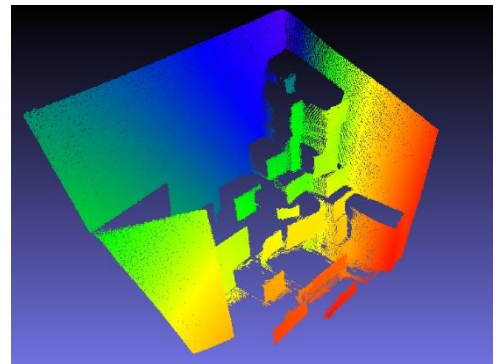
Flying Pixel Disabled



Flying Pixel Enabled



Flying Pixel Disabled (Point Cloud)



Flying Pixel Enabled (Point Cloud)

Get Temperature Data

- Temperature control provides the temperature readings of the sensor die, laser board, and module board in degrees Celsius (°C)

After selecting the **Get Temperature Data** user can see the options as shown below.

```

0 - Exit
1 - Back
2 - Main Menu
3 - get Laserboard Temperature
4 - get Moduleboard Temperature
5 - get Sensor die Temperature

Pick a Relevant Option: |
```

Get Laserboard Temperature used to get the Laserboard Temperature.

```
Pick a Relevant Option:      3
Laser Board Temperature data: 32
```

Get Baseboard Temperature used to get the Baseboard Temperature.

```
Pick a Relevant Option:      4
Moudle Board Temperature data: 36
```

Get Sensor Die Temperature used to get the Sensor Die Temperature.

```
Pick a Relevant Option:      5
Sensor Die Temperature data: 36
```

Post Processing

After selecting the **Post Processing** option, user can see the options as shows below.

```
0 - Exit
1 - Back
2 - Main
3 - Planarization
4 - Temporal Filter
5 - Undistort depth
6 - Depth Spatial Filter

Pick a Relevant Option:
```

Planarization:

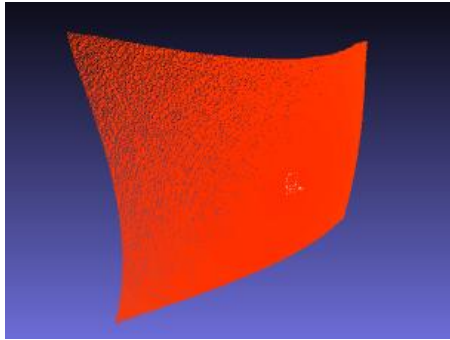
Planarization converts the radial (diagonal) depth into orthogonal real-world depth along the camera Z-axis.

After selecting the **Planarization** option, user can see the options as shown below.

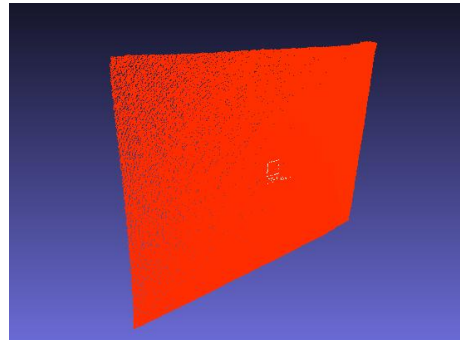
```
0 - Exit
1 - Back
2 - Main Menu
3 - Planarization OFF
4 - Planarization ON

Pick a Relevant Option: |
```

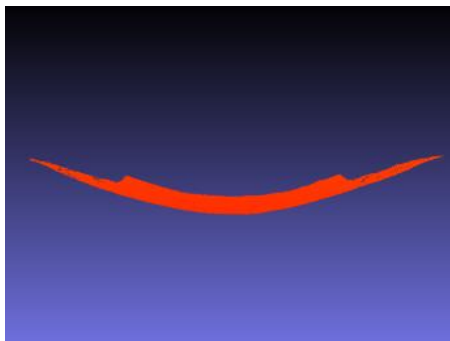
- **Planarization OFF** – used to disable the planarization filter.
- **Planarization ON** – used to enable the planarization filter.



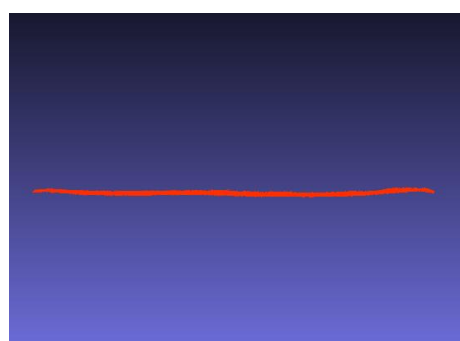
Planarization Disabled (Point Cloud)



Planarization Enabled (Point Cloud)



Planarization Disabled (Top View)



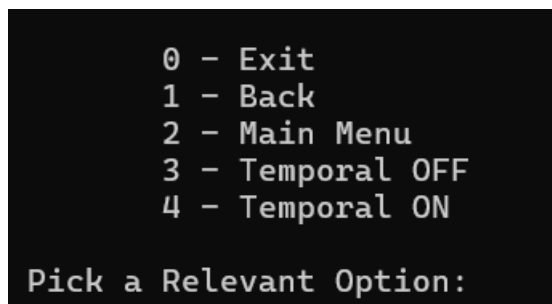
Planarization Enabled (Top View)

Temporal Filter:

DepthVista uses Adaptive Mean Temporal Filter which computes temporal variance over a sequence of frames and applies mean averaging when the variance is below a defined threshold.

This approach preserves sharp transitions and dynamic changes while still reducing noise in stable regions. As a result, it provides better edge and motion preservation compared to other temporal filters such as median and IIR filters

After selecting the **Temporal Filter** option, user can see the options as shown below.

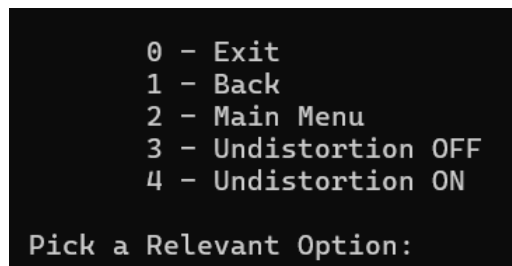


- **Temporal OFF** – used to disable the temporal filter.
- **Temporal ON** – used to enable the temporal filter.

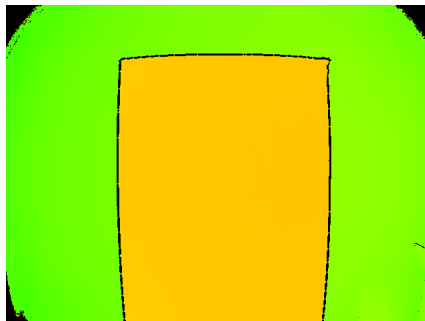
Undistort Depth:

Undistortion corrects optical distortions introduced by a camera lens so that straight lines and object shapes in the real world appear geometrically accurate in the image

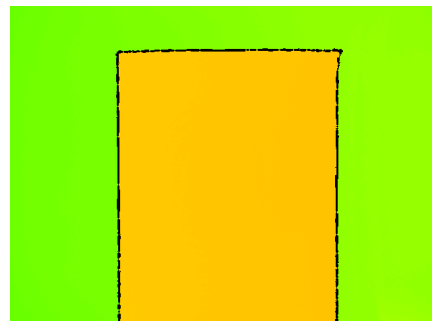
After selecting the **Undistort Depth** option, user can see the options as shown below.



- **Undistortion OFF** – used to disable the undistortion filter.
- **Undistortion ON** – used to enable the undistortion filter



Undistortion Disabled

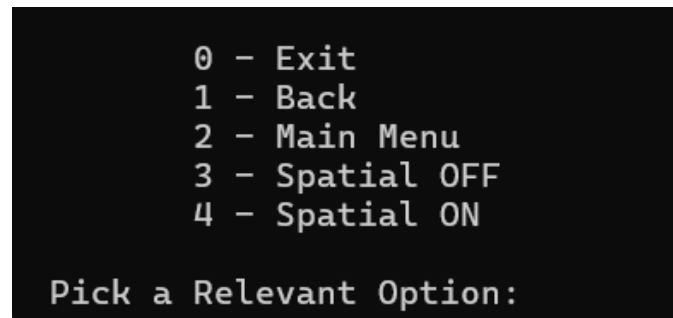


Undistortion Enabled

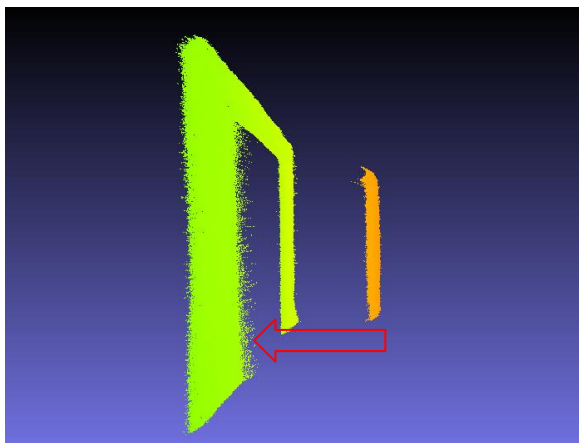
Depth Spatial Filter

Spatial filter reduces pixel-level variance across local neighborhoods. DepthVista uses Bilateral Spatial filter which is highly effective in reducing depth variance while preserving edge details. The following figure compares the depth frame before and after applying spatial filter.

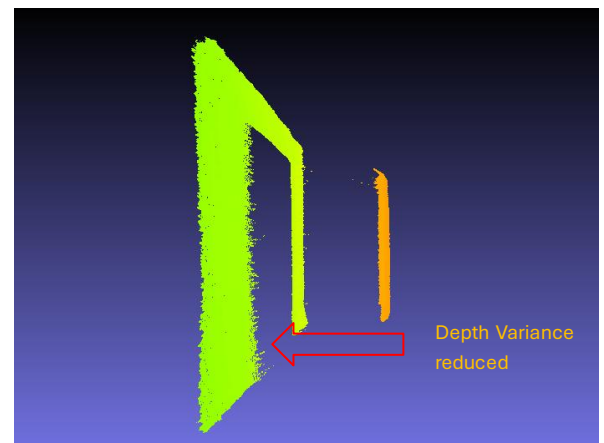
After selecting the **Depth Spatial Filter** option, user can see the options as shown below.



- **Spatial OFF** – used to disable the spatial filter.
- **Spatial ON** – used to enable the spatial filter.



Depth Spatial Disabled



Depth Spatial Enabled

Confidence Mode

User can select between Amplitude or SNR-based confidence frame using this control.

After selecting the **Confidence Mode** user can see the options as shown below.

```
0 - Exit
1 - Back
2 - Main Menu
3 - Set Confidence Mode
4 - Get Confidence Mode

Pick a Relevant Option:
```

Set Confidence Mode

After selecting the **Set Confidence Mode** user can see the options as shown below.

```
0 - Exit
1 - Back
2 - Main Menu
3 - Amplitude Frame
4 - SNR Frame

Pick a Relevant Option :
```

- **Amplitude Frame** - used to set the Amplitude Frame.
- **SNR Frame** – used to set the SNR Frame.

Get Confidence Mode

After selecting the **Get Confidence Mode** user can see the current confidence mode of the device as shown below.

```
Pick a Relevant Option:      4
Confidence Mode of the camera 0 : Amplitude Frame
```

Avg Depth

After selecting the **Avg Depth** option user can see the options as shown below.

```
0 - Exit
1 - Back
2 - Main Menu
3 - Centre
4 - Custom co-ordinate

Pick a Relevant Depth value position:
```

1. **Centre** option used get the values form centre of the preview.


```
Enter width within range(4 to 200) of average kernal for camera 0 : 32
Enter height within range(4 to 200) of average kernal for camera 0 : 32
Average Depth Value of camera 0 : 2600
Standard Depth Value of camera 0 : 33
Average IR Value of camera 0 : 23
Standard IR Value of camera 0 : 4
```

2. **Custom Co-Ordinate** option used to get the values form the any position of preview.

```
Enter width range (4 to 200) of average kernal for camera 0 : 32
Enter height range (4 to 200) of average kernal for camera 0 : 32
Enter X co-ordinate between the range 17 to 623 for camera 0: 234
Enter Y co-ordinate between the range 17 to 463 for camera 0: 213
Average Depth Value of camera 0 : 2685
Standard Depth Value of camera 0 : 33
Average IR Value of camera 0 : 24
Standard IR Value of camera 0 : 4
```

Unique ID

Unique ID option used to get the unique id for the current device.

```
Unique ID of the Camera 0 is 3104671244352620800
```

Read Firmware Version

Read Firmware Version option used to get the firmware version for the current device.

```
Firmware Version in camera 0 : 5.1.387.5
```

Serial Number

Serial Number option used to get the serial Number for the current device.

```
0 - Exit
1 - Back
2 - Main Menu
3 - Get Serial Number
```

```
Pick a Relevant Option:
```

Get Serial Number will display the serial Number of the current device as below

```
Pick a Relevant Option:      3
Serial Number for the camera index 0 is : E1-PROL251144
```

Laser Safety

To comply with Class 1 laser safety requirements defined in IEC 60825 for eye safety, the maximum Accessible Emission (AE), or optical laser power, must remain below the Accessible Emission Limit (AEL). The maximum optical peak power of the lasers depends on the streaming mode, maximum allowed integration time, and laser duty cycle.

When laser safety monitoring is enabled, the system continuously monitors laser activity to ensure compliance with Class 1 laser safety requirements under all operating modes, including abnormal conditions and system malfunctions.

If any abnormal emission condition is detected, the system shuts down the VCSELs to ensure user safety, until further command.

Note: Laser Reset is currently not supported. To re-enable the VCSELs, either perform a power reset or disable laser safety monitoring.

Laser safety option is used to set and get the laser safety feature for the current device.

```
0 - Exit
1 - Back
2 - Main Menu
3 - Set Laser Safety Mode
4 - Get Laser Safety Mode
5 - Laser Reset
```

```
Pick a Relevant Option:
```

Selecting the **Set Laser Safety Mode** will display the option to Disable/Enable the laser safety as follows.

```
Pick a Relevant Option:      3

0 - Exit
1 - Back
2 - Main Menu
3 - Disable Laser Safety
4 - Enable Laser Safety
```

Selecting the **Get Laser Safety Mode** will show the current mode of the device.

```
Laser Safety of the camera 0 :  ON
```

```
Laser Safety of the camera 0 :  OFF
```

FPS Control

FPS Control option is used to set and get the FPS for the current device.

```
0 - Exit
1 - Back
2 - Main Menu
3 - Set FPS Mode
4 - Get FPS Mode
```

```
Pick a Relevant Option:
```

Selecting the **Set FPS Mode** will allow the user to set the FPS for the current device will list the menu as below

```
0 - Exit
1 - Back
2 - Main Menu
3 - 30 FPS
4 - 50 FPS
```

```
Pick a Relevant Option:
```

User can select 30 or 50 FPS for the current device and the message for successful FPS set will be displayed as below

```
Current FPS Mode : 30 FPS
FPS mode is successfully set in camera 0 with FPS 30
```

```
Current FPS Mode : 30 FPS
FPS mode is successfully set in camera 0 with FPS 50
```

Note: 50 FPS mode for HD data mode is not supported for DepthVista Helix USB variant.

Selecting the **Get FPS Mode** will allow the user to get the FPS for the current device as below

```
Pick a Relevant Option:      4
Current FPS Mode of the camera 0 : 30 FPS
```

```
Pick a Relevant Option:      4
Current FPS Mode of the camera 0 : 50 FPS
```

Troubleshooting

In the DepthVista sample application, the device is selected but the preview window is black.

- Please make sure that the external power supply is connected to the device and then restart the application.
- Please check whether the device is connected to USB 2.0. If so, as this device supports only USB 3.2 Gen 1 interface, please connect the device to USB 3.2 Gen 1 port and then restart the application.
- You need to install the latest version of DepthVista sample application from the Developer Resources website.

Make sure external power supply is connected, and the device is connected to USB 3.2 Gen 1 Interface. Then in the DepthVista sample application, the preview window is black.

It seems like no image is received from the camera. Contact e-con Systems online support support@e-consystems.com.

Error: error while loading shared libraries: libdc1394.so.25: cannot open shared object file: No such file or directory

Warning: libdc1394.so.25, needed by /usr/lib/libopencv_world.so.4.2.0, not found

Run the following command in terminal to solve the issue.

```
sudo apt-get install libdc1394-25
```

Support

Contact Us

If you need any support on DepthVista product, please contact us using the Live Chat option available on our website - <https://www.e-consystems.com/>

Creating a Ticket

If you need to create a ticket for any type of issue, please visit the ticketing page on our website - <https://www.e-consystems.com/create-ticket.asp>

RMA

To know about our Return Material Authorization (RMA) policy, please visit the RMA Policy page on our website - <https://www.e-consystems.com/RMA-Policy.asp>

General Product Warranty Terms

To know about our General Product Warranty Terms, please visit the General Warranty Terms page on our website - <https://www.e-consystems.com/warranty.asp>

Revision History

Rev	Date	Description	Author
1.0	05-Jan-2026	Initial Draft	Camera Products
1.1	12-Feb-26	Changed device introduction and descriptions for some controls	Camera Products
1.2	25-March-26	Added new Confidence control and post processing filters	Camera Products
1.3	28-May-26	Added new controls – SerialNO, Laser Safety and FPS Mode. Added sample images for post processing filters.	Camera Products